

2 (a) Cotton bollworms, *Helicoverpa* spp., are insect pests of cotton.

Adult cotton bollworms are moths. The adult female moths lay eggs on cotton plants. The eggs hatch into larvae. The larvae feed on cotton plants, causing extensive damage and reduction in yield.

Fig. 2.1 shows a mature cotton fruit (cotton boll) from an uninfested plant.

Fig. 2.2 shows a cotton bollworm larva inside a damaged cotton boll.



Fig. 2.1

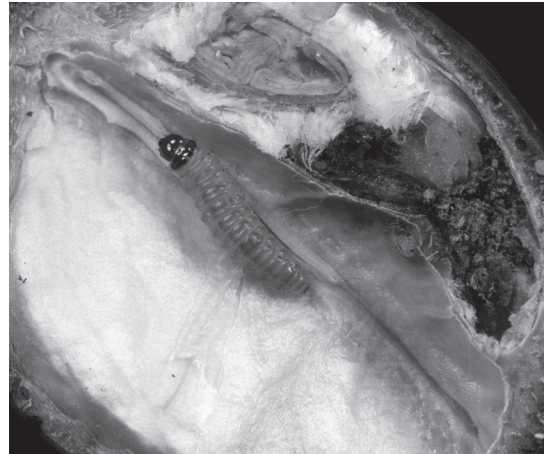


Fig. 2.2

A gene, *cry1Ac*, from the bacterium *Bacillus thuringiensis* (Bt), can be inserted into the cotton genome to produce Bt cotton.

- The protein, Cry1Ac, coded by *cry1Ac* is toxic to some species of bollworm.
- This toxicity gives cotton plants some resistance to cotton bollworm.

Studies of the effectiveness of the protein Cry1Ac showed that older plants produced less of the protein. As a result, spraying with insecticide was still needed when larval density increased.

An improved Bt cotton with two genes, *cry1Ac* and *cry2Ab*, produced two different proteins and was expected to give protection from *Helicoverpa* spp. for the whole growing season.

This improved Bt cotton was introduced into Australia in 2004. An investigation was carried out to determine whether this improved Bt cotton would also need spraying with insecticide.

Two different farms were used. In each farm, a standard-sized plot at the centre of a cotton field was marked out.

- Each plot measured 20 m².
- Each plot was divided into 3 sets of 8 rows of cotton.
- Each row of cotton was divided into 1 metre sections, each with 8–10 plants.

Two treatments were used in each plot in a random pattern.

- In treatment 1, the researchers removed all the larvae from the plants.
- In treatment 2, the researchers made sure that there were 3 medium-sized larvae (9–16 mm in length) per metre section.

Fig. 2.3 shows the arrangement of a 20 m² plot in a cotton field.

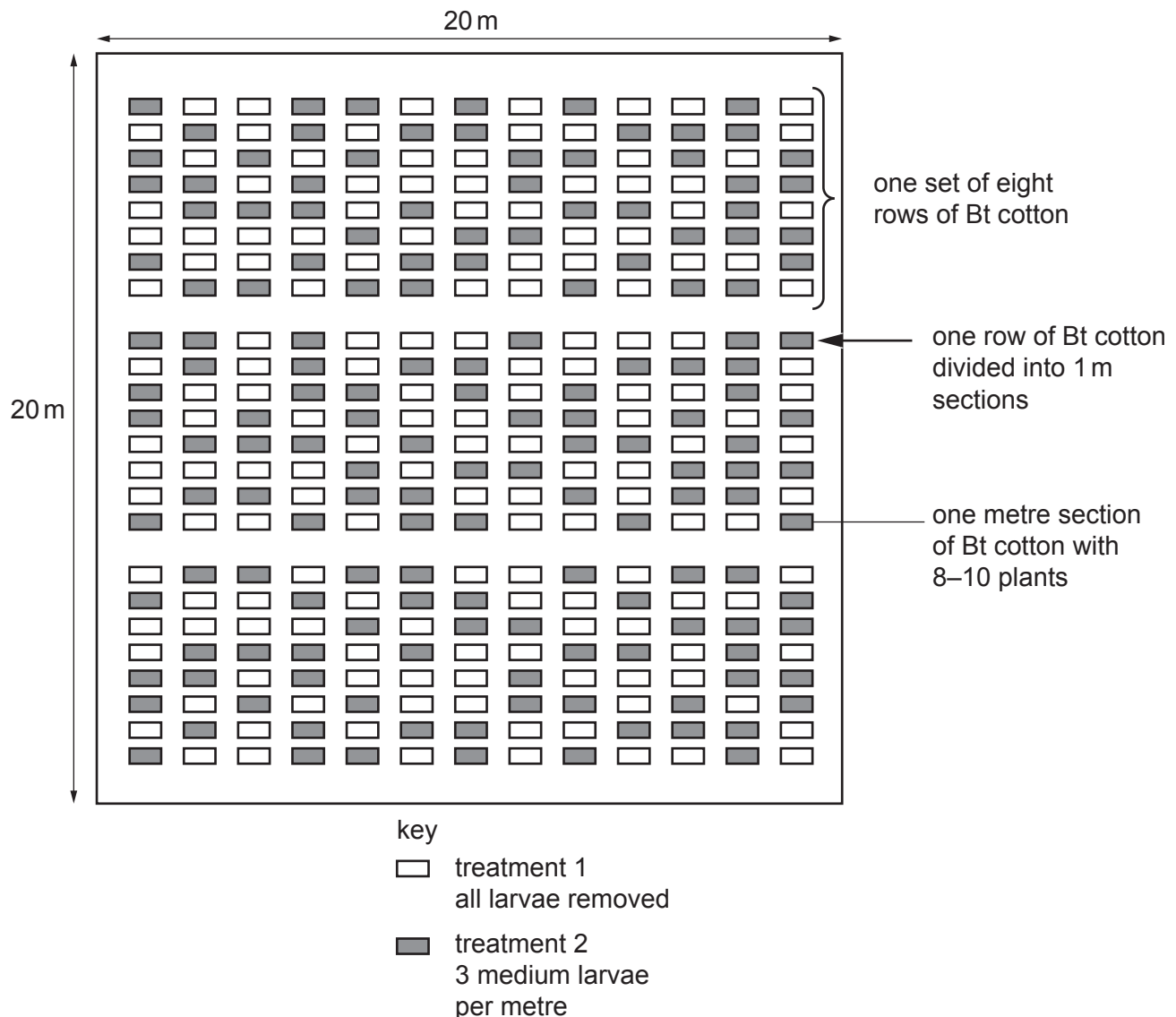


Fig. 2.3

- After three days, the plot on one farm was sprayed with an insecticide and the plot on the other farm was not sprayed with an insecticide.
- The cotton bolls were picked at the same time, when all the bolls in each plot were open.
- The cotton bolls were weighed.
- The cotton fibres were separated from the cotton bolls and weighed.

(i) Suggest how the researchers made sure that there were 3 medium-sized larvae in each section for treatment 2.

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(ii) Identify **two** variables that have been standardised in this investigation.

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(iii) Explain why one of the plots was **not** sprayed with insecticide.

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(b) Table 2.1 shows the results of this study.

Table 2.1

plot	not sprayed with insecticide		sprayed with insecticide	
treatment	1 (larvae removed)	2 (3 larvae present)	1 (larvae removed)	2 (3 larvae present)
mean mass of cotton bolls per section /g $\pm$ s	521.2 $\pm$ 77.6	418.6 $\pm$ 74.2	615.5 $\pm$ 92.5	605.6 $\pm$ 118.1
mean mass of cotton fibre per section /g $\pm$ s	223.8 $\pm$ 30.9	203.7 $\pm$ 23.5	251.3 $\pm$ 38.6	240.0 $\pm$ 56.0

A *t*-test can be used to find out if treatment 2 has any effect on the yield of cotton fibre.

(i) State **one** reason why the *t*-test is suitable for the data in Table 2.1.

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(ii) State a null hypothesis for the *t*-test to compare the effect of insecticide on yield of cotton fibre when larvae are present.

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- (c) The researchers were trying to find out if farmers growing improved Bt cotton will benefit from spraying the improved Bt cotton with insecticide to control cotton bollworm infestations.

Comment on whether the results in Table 2.1 show that improved Bt cotton should be sprayed with insecticide.

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[Total: 9]